

Client Server Model

Client

Anything (device or software) the user operates is called client

Anything User Use Karta Hai and wusse internet chalaata Hai

User = client

Server

Duniya mai Rakha hai Computer programmed hai Serve karne ke liye

The Computer is programmed to Serve the Service to the requester.

Client Request Send karta hai
Server Respond karta Hai

Server = Computer/Machine where website's code, database, files are stored

Server request dekhta hai and Response detta hai

Server is a programmed computer to Serve requests of Client.

HTTP Request Response cycle

Hypertext Transfer Protocol

HTTP protocol have Rules for Sending & Recieving data

Rules around Request Response cycle

Request, processing, Response, Render

Jaise Tumne Zomato pe Pizza order kiya (request)
Restaurant tumhara order prepare karta hai (processing)
fir pizza delivery Boy tumhe pizza deliver karta hai (Response)
fir Tum pizza dabba kholke khathe ho (render)

What Happens When you Visit a website

When you type `www.example.com`

- ① Browser checks cache → Pehle dekh lega ki koi Saved copy hai ya nahi

If the website already visited then some data is stored related to that particular website in the cache to decrease the work & loading time of the website

If cache dont have data then DNS lookup

- ② DNS lookup

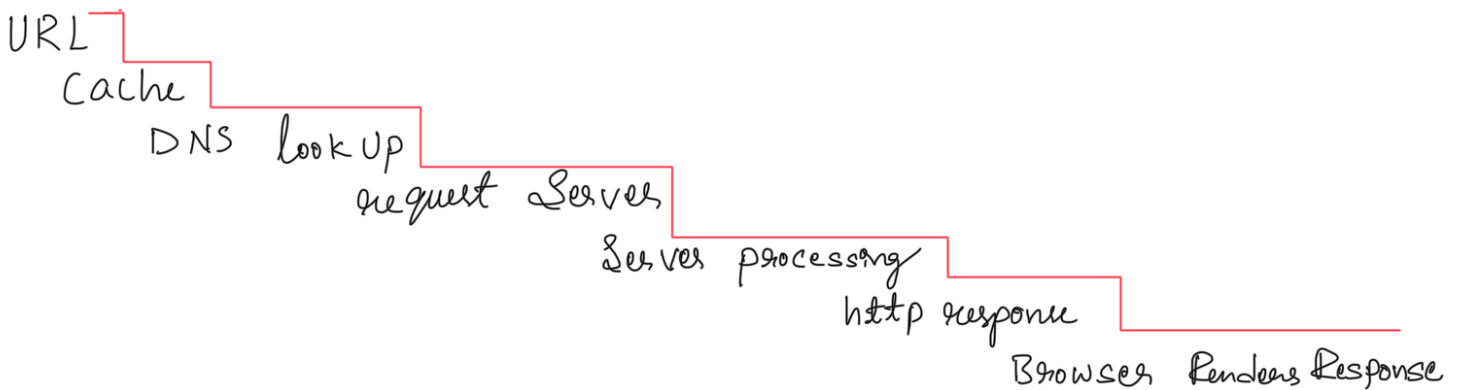
Domain Name (Example.com) ko IP Address me Convert karta Hai

③ Browser Sends HTTP Request to Server

Server ko bolta hai "Mujhe Homepage Bhejo"

④ HTTP Response Back to Browser

Data :-> packets ke form me HTML, CSS, JS
Jetha Hai.



Difference Between frontEnd and BackEnd

Front End

ye wo part jo user dekh sakta hai aur interact karta hai -> Buttons, form, text images.

Back End

ye wo part hai jo tumhe dikhta nahi hai par hota hai -> data store karna, logic handle karna authentication, Payment etc.

Static and Dynamic Web Pages

Static Websites

- Has User ko Ek Same content dikhata hai
- Sirf HTML, CSS Se Ban Sakhta hai
- Fast Simple Cheap hosting
- Example : old portfolio pages, Wikipedia, Quora

Dynamic Website

- Has User ko Customized / live content detha hai
- Back End + DataBase ki help Se Banta hai
- Content change hota hai User ke hisab Se
- Ex : Facebook, youtube, E-commerce Homepages

Web Hosting is a Service

Web Hosting = Service where your website lives on a Server so that anyone in the world can access it.

Hosting companies like (AWS, Netlify, Heroku) rent your space on their Server.

Process

1. Upload your website files to a hosting Server
2. Server stays 24x7 Active Up and Running
3. When Someone types your domain, request goes to that hosting Server
4. Server delivers files / data → Website loads on their browser.